

Yousef Mohammad

Software Engineer | Full-Stack, Backend, Applied AI, Data-Driven Systems

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Ships Real Systems

AWAITS + InKuiS

Practical AI

OpenAI + RAG + Search

Broad Software Base

Nuxt/Vue + Docker + CI/CD

Professional Profile

Final-year M.Sc. Software Engineering student and Research Engineer with practical experience shipping web applications, APIs, AI-assisted workflows, dashboards, and deployment pipelines at Linnaeus University. Strongest work includes AWAITS, InKuiS, and visual-analytics systems built with Nuxt/Vue, TypeScript, Python, MongoDB, OpenAI APIs, Docker, GitLab/GitHub CI/CD, and data visualisation tooling. Open to software engineering, full-stack, backend, applied AI, data, and junior/platform roles.

Technical Skills

Languages	TypeScript, JavaScript, Python, Java, SQL, C++, Dart, GLSL
Frontend	Vue 3, Nuxt 3, React, Flutter, Vuetify, Tailwind CSS, HTML/CSS, D3.js, Chart.js, Three.js
Backend & APIs	Node/Nitro server APIs, Flask, FastAPI, REST APIs, authentication, RBAC, SSE streaming, geocoding APIs
Data & AI	MongoDB, PostgreSQL, MySQL, OpenAI API, Assistants API, RAG, vector stores, semantic search, prompt orchestration, BioBERT, PubMedCLIP, PyTorch, TensorFlow
DevOps & Tools	Docker, Docker Compose/Swarm, Git, GitHub, GitLab CI/CD, Kaniko, Nginx, AWS S3, Cloudinary, Prefect, Linux, Vitest, MLflow

Experience

Research Engineer / Full-Stack Developer

Linnaeus University

Jun 2024 – Present

Växjö, Sweden

- Built and maintained AI-enabled research software across AWAITS, InKuiS, and visual-analytics projects, contributing to frontend, backend APIs, database models, authentication, AI integration, deployment, and user-facing workflows.
- Developed production-oriented Nuxt 3/Vue applications with TypeScript, MongoDB, OpenAI APIs, role-based access control, dashboards, analytics, file handling, and responsive interfaces.
- Implemented Docker-based deployments and CI/CD workflows using GitHub/GitLab pipelines, Kaniko image builds, Docker Swarm/Compose, SSH deployment, health checks, and production configuration management.
- Worked with researchers and stakeholders to translate research ideas into usable prototypes and deployed systems, iterating from requirements to implementation, testing, and release.

Evening Shift Manager

ICA Supermarket Mörnars

2013 – Present

Växjö, Sweden

- Lead evening operations, coordinate staff, handle customer and operational escalations, and maintain service quality in a high-pressure retail environment.

Selected Practical Projects

AWAITS – AI Writing Feedback Platform

Nuxt 3, TypeScript, MongoDB, OpenAI, Docker, GitLab CI/CD

- Developed a deployed educational platform for AI-assisted academic writing feedback: rich-text editing with TipTap, AI inline suggestions, chat/review flows, teacher dashboards, user management, consent handling, and analytics.
- Implemented topic-based architecture connecting OpenAI Assistants, vector stores, MongoDB topic records, owners, enrolled students, and role-based permissions with cleanup logic for partial failures.
- Built streaming feedback with Server-Sent Events, OpenAI queue/rate-limit handling, prompt orchestration, file-management APIs, external web-search support, and feedback statistics.
- Supported production delivery with Docker, GitLab CI/CD, Kaniko image builds, Docker Swarm deployment, health checks, deployment notifications, and Locust load testing with zero failed requests in recorded tests.

InKuiS – Event Discovery and Management Platform

Nuxt 3, TypeScript, MongoDB, OpenAI, Prefect, S3/Cloudinary

- Built full-stack features for a deployed event platform with event CRUD, authentication, user profiles, image upload, attendee flows, search/filtering, calendar views, analytics dashboards, and bilingual English/Swedish UI.
- Implemented location-aware event discovery using OpenStreetMap/Nominatim geocoding, GPS/reverse geocoding, radius filtering from 1–100 km, proximity sorting, and distance display.
- Integrated an OpenAI-powered chatbot that converts natural-language event questions into structured MongoDB queries with

keyword extraction, date/category/location/price filters, and streaming responses.

- Built Prefect workflows that fetch uncategorized events, classify them with OpenAI in batches, validate categories, update public API endpoints, and deploy flows through GitLab CI/CD.

Medical Case Visual Analytics Platform

Nuxt/Vue, D3.js, FastAPI, BioBERT, PubMedCLIP

- Built a full-stack analytics tool for exploring 7,400+ medical cases and 40,900+ images using UMAP views, linked filters, grouping/comparison mode, dashboards, and case galleries.
- Added multimodal similarity search using BioBERT text embeddings and PubMedCLIP image embeddings, allowing query-based and example-image case retrieval.

Multi-Target Compiler for OFP Language

Java, ANTLR, ASM, JVM bytecode

- Implemented lexer/parser grammar, symbol tables, scope management, semantic analysis, type checking, Python code generation, and JVM bytecode generation for a custom imperative language.
- Supported functions, recursion, arrays, control flow, and multiple primitive types including int, float, bool, char, and string.

AI Impact 2030 and Demographics Dashboards

JavaScript, D3.js, Data Visualisation

- Built interactive dashboards with linked scatter plots, parallel coordinates, histograms, box plots, brushing, lasso/marquee selection, coordinated filtering, and data-driven storytelling.

Education, Research & Publication

M.Sc. in Engineering, Software Engineering

2021 – 2026 expected

Linnaeus University

Växjö, Sweden

- **Master's thesis:** RiskRank – context-aware vulnerability prioritization for CI/CD using scanner findings, fan-in ranking, git-history signals, LLM reachability analysis, and reviewer-facing remediation reports.
- **Bachelor thesis:** Built and evaluated a RAG-based academic-writing tutor using OpenAI Assistants, retrieval, vector-store grounding, and comparative evaluation against non-RAG behaviour.
- **Publication:** Co-authored *AWAITS*, a refereed conference paper on an AI-supported academic writing tutoring system for cognitive skills and AI-assisted learning.

Additional Projects & Exposure

Misty robotics / LLM conversation system: Prompt orchestration, dialogue-state management, tool calls, output parsing, and HTTP/WebSocket-based robot interaction.

Data-intensive and ML systems: Flask/React health-check app, Dockerized backend, notebooks, AIMO/weakest-link UI, pose-analysis frontend, and render.com deployment workflow.

Graphics and GPU programming: Python OpenGL/SDL2 museum walkthrough with shaders and real-time camera controls; CUDA matrix multiplication and memory-access optimisation.

Languages

English (professional) | Swedish (native) | Arabic (native)